

Pre-Registered Fermion Mass Predictions

for comparison with PDG 2026

Andrew M. Brilliant

Applied Dynamics Research, Sapporo, Japan

ab@ad-research.org

Date: May 18, 2026

Single input: $\mu^* = m_e + m_\mu + m_\tau = 1883.099 \text{ MeV}$

Derived offset: $\delta = 2/9$ (Shulga, arXiv:2605.10245)

Cascade constant: $\alpha_K = \sqrt{3/2} - 1$, $R_K = 1/\alpha_K^2 = 19.798$

Particle	Closed-form expression	Predicted (MeV)	Scale
tau	$(\mu^*/6)(1+\sqrt{2} \cos(2/9))^2$	1776.93	pole
mu	$(\mu^*/6)(1+\sqrt{2} \cos(2/9+4\pi/3))^2$	105.658	pole
e	$(\mu^*/6)(1+\sqrt{2} \cos(2/9+2\pi/3))^2$	0.51100	pole
s	$\alpha_K^2 * \mu^*$	95.12	$\overline{\text{MS}}(\mu^*)$
d	$\alpha_K^4 * \mu^*$	4.804	$\overline{\text{MS}}(\mu^*)$
u	$\alpha_K^2 * \sqrt{2m_e * \mu^*}$	2.216	$\overline{\text{MS}}(\mu^*)$

Self-consistency test (4 equations, 3 unknowns):

Test	Path 1	Path 2	Agreement
m_s dual	$F^2(e, \mu, \tau) = 95.12$	Sum rule + cascade = 95.17	0.058%
m_d cross-check	Cascade: 4.804 MeV	Tau-slice $F^2(Z, e, u) = 4.865$	0.49%
m_u prediction	Cascade: 2.216 MeV	PDG: $2.20 \pm 0.07 \text{ MeV}$	+0.23 sigma

Systematic: QED corrections for charge-2/3 quarks ~1%, giving $\pm 0.02 \text{ MeV}$ on m_u . With this: $m_u = 2.22 \pm 0.02 \text{ MeV}$ (approx 3x tighter than PDG).

What would falsify:

Prediction	Current dev.	Shift to 1-sigma	Tests
$m_s(\mu^*) = 95.12 \text{ MeV}$	+0.12 sigma	0.70 MeV shift	F-identity

$m_d(\mu^*) = 4.804 \text{ MeV}$	+0.10 sigma	0.17 MeV shift	Cascade
$m_u(\mu^*) = 2.216 \text{ MeV}$	+0.23 sigma	0.065 MeV shift	GM bridge

Generated 2026-05-18 10:41 UTC. Predictions frozen before PDG 2026 release. Zero continuous adjustable parameters once μ^* , $\delta=2/9$, and pair-normalized closure are fixed.

Full manuscript: DOI:10.20944/preprints202604.0693.v2